Our preferred choice:

Client Role & Context
We are AquaWatch Netherlands, an environmental consultancy advising Dutch water boards on the relationship between land use, agricultural activity, and surface water quality. With increasing pressure from EU regulations such as the Water Framework Directive, our clients need evidence-based insights into which regions face the greatest risk of water quality degradation and what land use factors are driving it. Our goal is to identify pollution hotspots and link them to specific upstream land use patterns to support targeted intervention planning.

Research Questions
RQ1. Which surface water monitoring locations in the Netherlands consistently report elevated levels of nitrates and phosphates, and what types of agricultural or industrial land use are dominant in their surrounding areas?
RQ2. Is there a measurable relationship between the density of livestock farming in a municipality and the water quality indicators measured at nearby monitoring stations?
RQ3. Which municipalities that overlap with Natura 2000 protected zones show the worst water quality trends, suggesting that current land use practices may be threatening protected ecosystems?

Datasets

Waterkwaliteitsportaal (Water Quality Portal) The national water quality database maintained by Rijkswaterstaat and the water boards, containing measurements of chemical and ecological indicators (nitrates, phosphates, heavy metals) at monitoring stations across the Netherlands. 🔗 https://www.waterkwaliteitsportaal.nl/

Natura 2000 – PDOK The official spatial dataset of all 162 Natura 2000 protected nature areas in the Netherlands, maintained by the Ministry of Agriculture, Nature and Food Quality and distributed via PDOK. Used to identify which monitoring stations fall near or within protected zones. 🔗 https://www.pdok.nl/introductie/-/article/natura-2000

CBS StatLine – Agriculture; crops, livestock and land use by region A CBS open dataset providing per-municipality data on livestock counts (cattle, pigs, poultry), crop types, and agricultural land use. Available via OData API and downloadable as CSV. 🔗 https://opendata.cbs.nl/statline/#/CBS/en/dataset/80783eng/table

Alternative:

Client Role & Context
We are GreenGrid Consulting, an advisory firm working with regional energy authorities in the Netherlands to support the national transition to renewable energy. Our mandate is to identify the most suitable locations for new wind and solar installations, balancing energy yield potential

with environmental protection and infrastructure constraints. Evidence-based site recommendations are essential to guide investment decisions and regulatory approval processes.

Research Questions
RQ1. Which municipalities in the Netherlands have the highest potential for solar energy installation, when considering solar irradiance levels, available land use types (e.g., agricultural, industrial), and proximity to the existing electricity grid?
RQ2. Which areas are suitable for wind energy development, and which of those areas overlap with protected nature reserves or Natura 2000 zones, making them unsuitable despite high wind potential?
RQ3. When combining wind and solar potential, which regions offer the best opportunity for hybrid renewable energy zones that are both high-yield and free of environmental or infrastructural conflicts?

1. Global Solar Atlas
Provides global raster data on photovoltaic power potential and solar irradiance at high resolution. Can be used to extract per-municipality solar yield estimates for the Netherlands.
🔗 https://globalsolaratlas.info/download
2. Global Wind Atlas
Provides wind speed and wind power density data at multiple hub heights globally, including at municipality level. Used to assess wind energy potential across Dutch regions.
🔗 https://globalwindatlas.info/en/download/gis-files

3. PDOK – Dutch Land Use Dataset (Basisregistratie Gewaspercelen / CBS Bodemgebruik)
A Dutch government dataset classifying land parcels by use type (agricultural, industrial, nature, urban, etc.). Critical for filtering locations that are physically available for installation.
🔗 https://www.pdok.nl/introductie/-/article/basisregistratie-gewaspercelen-brp-
(Last dataset seems best to me for this scope)